

Half Adder

Dataflow

```
1  module half_adder_df (  
2      input logic a, b,  
3      output logic sum, carry  
4  );  
5      assign sum = a ^ b;  
6      assign carry = a & b;  
7  endmodule
```

Behavioral

```
Behavioral > ≡ halfAdderBH.v  
1  module half_adder_bh (  
2      input logic a, b,  
3      output logic sum, carry  
4  );  
5      always_comb begin  
6          sum = a ^ b;  
7          carry = a & b;  
8      end  
9  endmodule
```

Structural

```
Structural > ≡ halfAdderST.v  
1  module half_adder_st (  
2      input logic a, b,  
3      output logic sum, carry  
4  );  
5      // Using built-in gate primitives (positional only)  
6      xor sum_gate (sum, a, b);  
7      and carry_gate (carry, a, b);  
8  endmodule  
9
```

Full Adder Dataflow

```
1  module full_adder_df (  
2      input logic a, b, c,  
3      output logic sum, carry  
4  );  
5      assign sum = (a ^ b) ^ c;  
6      assign carry = (a & b) ^ ((a ^ b) & c);  
7  endmodule
```

Behavioral

```
1  module full_adder_bh (  
2      input logic a, b, c,  
3      output logic sum, carry  
4  );  
5      always_comb begin  
6          sum = (a ^ b) ^ c;  
7          carry = (a & b) ^ ((a ^ b) & c);  
8      end  
9  endmodule
```

Structural

```
1  module full_adder_st (  
2      input logic a, b, c  
3      output logic sum, carry  
4  );  
5      wire temp1, temp2, temp3;  
6      // Using gate primitives  
7      xor x1(temp1, a, b);  
8      xor x2(sum, temp1, c);  
9  
10     and a1(temp2, a, b);  
11     and a2(temp3, temp1, c);  
12     xor x3(carry, temp3, temp2);  
13 endmodule
```

Pre Lab

1. What distinguishes combinational circuits from sequential circuits?

- Combinatorial circuits depend on the current input values to produce outputs, sequential circuits depend on both the current inputs and the previous state stored in memory elements to produce an output.

2. Derive the sum and carry expressions for half and full adders from their truth tables.

- Half Adder

A	B	$(A \oplus B)$ (Sum)	$(A \& B)$ (Carry)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

- Full Adder

A	B	C	$A \oplus B$	$(A \oplus B) \oplus C$ (Sum)	$(A \oplus B) \& C$	$A \& B$	$(A \& B) \& ((A \oplus B) \oplus C)$ (Carry)
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0
0	1	0	1	1	1	0	0
0	1	1	1	0	0	0	1
1	0	0	1	1	1	0	0
1	0	1	1	0	0	0	1
1	1	0	0	0	0	1	1
1	1	1	0	1	0	1	1

3. Derive the difference and borrow expressions for half and full subtractors.

- Half Subtractor

A	B	$\sim A$	$A \oplus B$ (Difference)	$\sim A \& B$ (Borrow)
0	0	1	0	0
0	1	1	1	1
1	0	0	1	0
1	1	0	0	0

- Full Subtractor

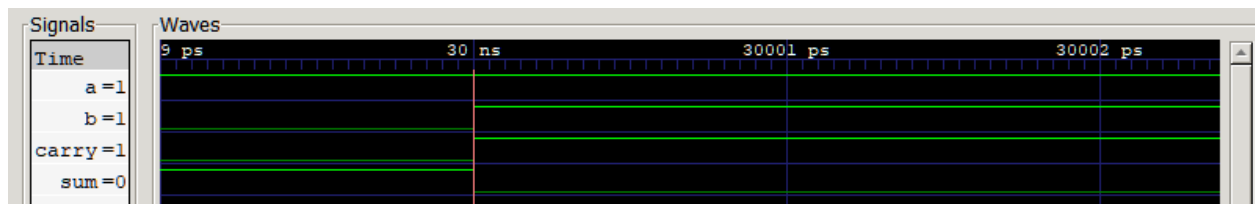
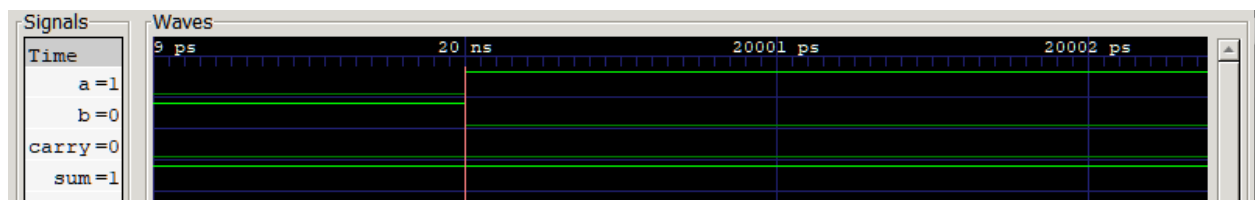
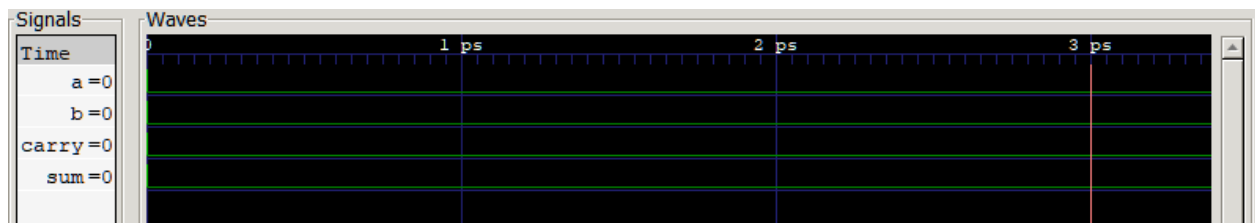
A	B	C	$A \oplus B$	$A \oplus B \oplus C$ (Difference)	$\sim A$	$\sim A \& B$	$\sim A \oplus B$	$\sim(A \oplus B) \& C$	$\sim(A \oplus B) \& C$ $\sim A \& B$ (Borrow)
0	0	0	0	0	1	0	1	0	0
0	0	1	0	1	1	0	1	1	1
0	1	0	1	1	1	1	0	0	1
0	1	1	1	0	1	1	0	0	1
1	0	0	1	1	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0
1	1	0	0	0	0	0	1	0	0
1	1	1	0	1	0	0	1	1	1

4. Explain the advantages of SystemVerilog's logic data type over traditional Verilog's reg and wire.

- SystemVerilog's logic simplifies code by merging the capabilities of wire and reg, reduces errors during refactoring, and improves readability—making it the preferred default for most single-driver signals in modern RTL design.

5. What is the purpose of the -g2012 flag in Icarus Verilog?
 - The -g2012 flag tells the compiler to enable IEEE 1800-2012 (SystemVerilog 2012) features.
6. What is the purpose of the \$dumpvars system task in Verilog testbenches?
 - The \$dumpvars system task in Verilog is used in testbenches to record signal activity into a Value Change Dump file for waveform viewing with tools like GTKWave.
7. What are the benefits of using named port connections (.port(name)) over positional connections?
 - Named port connections in Verilog (.port(signal)) are safer and easier to maintain than positional connections because they explicitly label each signal, improving readability and reducing errors. Unlike positional connections, they are unaffected by changes in port order, allow partial connections, and make refactoring or debugging simpler—especially in large or collaborative projects. While positional connections may be acceptable for very small, simple modules, named connections are considered best practice for most real-world designs.

Half Adder



Full Adder

